

LaserMOD Tutorial

EE232

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Outline

- Introduction to LaserMOD
- Walk-through of basic laser structure simulation
- Notes to look out for when using the program
- Simulation verification charts
- Useful formulas



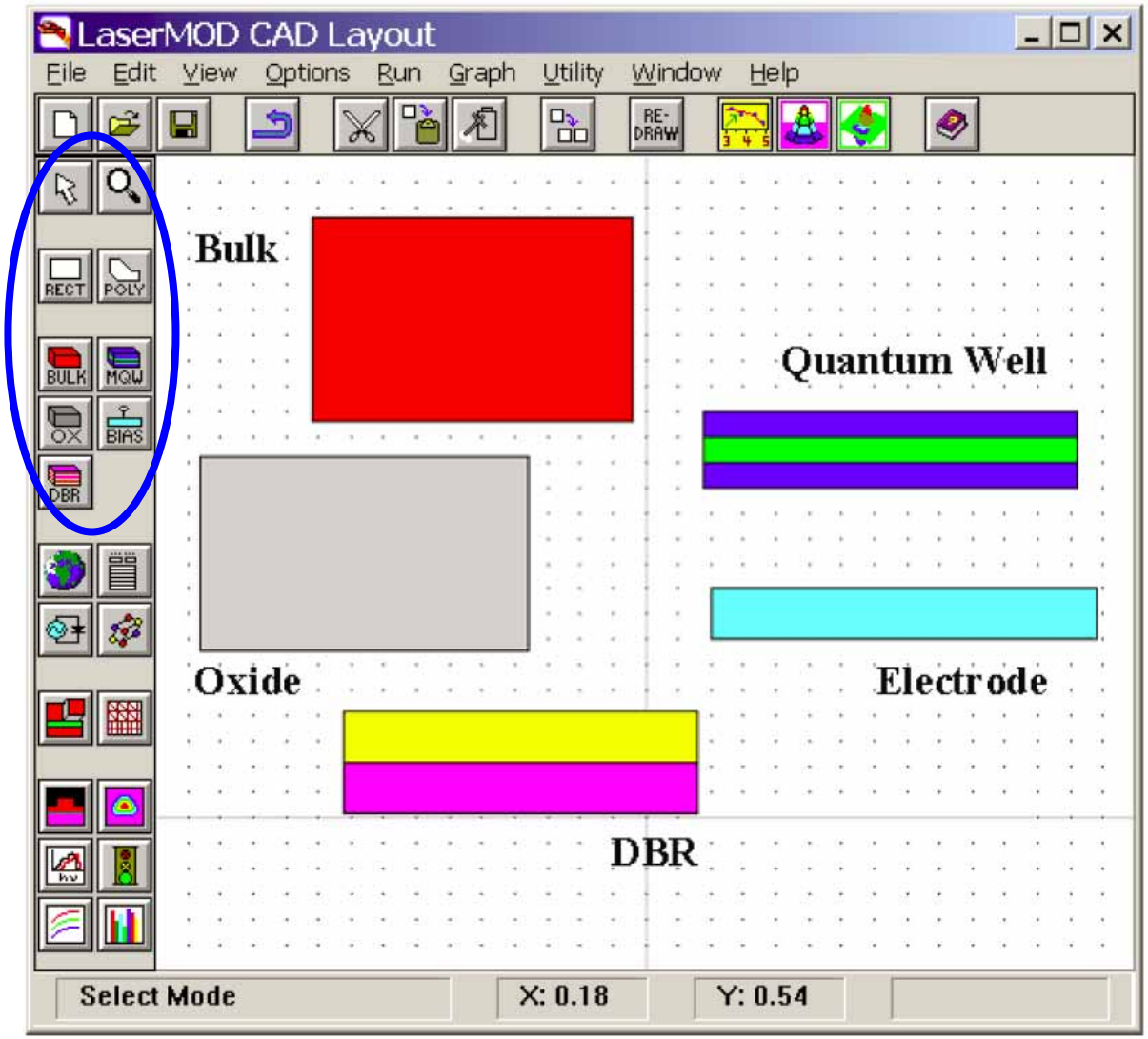
Simulation Overview

1) Build laser structure
   a) Draw structure
   b) Refine mesh
2) Set bias conditions (steady state or transient)
3) Simulate Index/Doping profiles
4) Calculate Gain
   a) Allows access to “peak gain” curve and other gain data
5) Simulate Laser
   a) Simulating a transient condition allows access to “frequency response” curve
6) Generate additional plots (be sure to verify “Plot Data for Bias #”)
   a) Standard plots
   b) Data vs. Bias (custom plots)
   c) Spatial Data (custom plots)



LaserMOD

These are the laser building tools





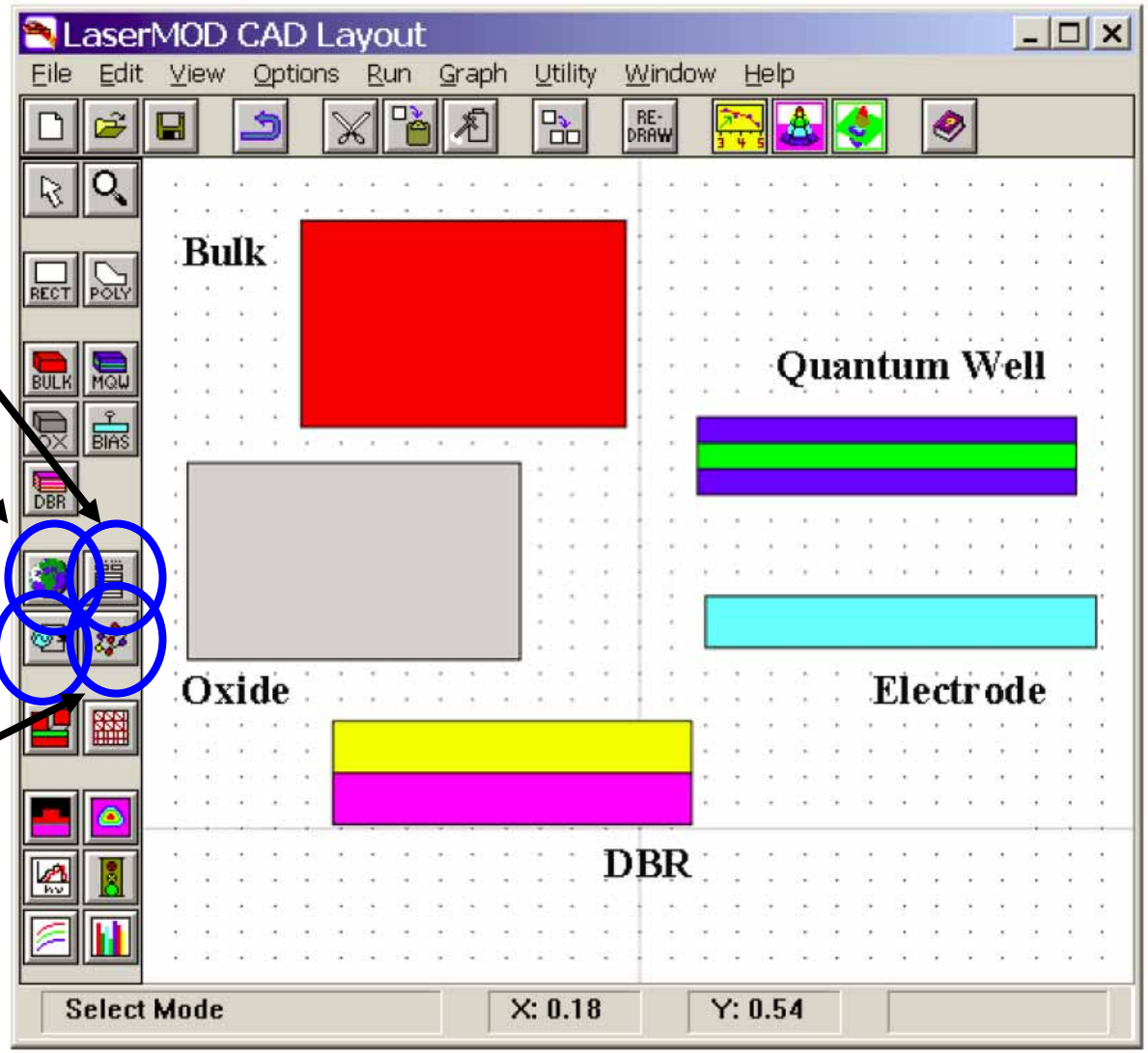
LaserMOD

Edit symbols

Global settings

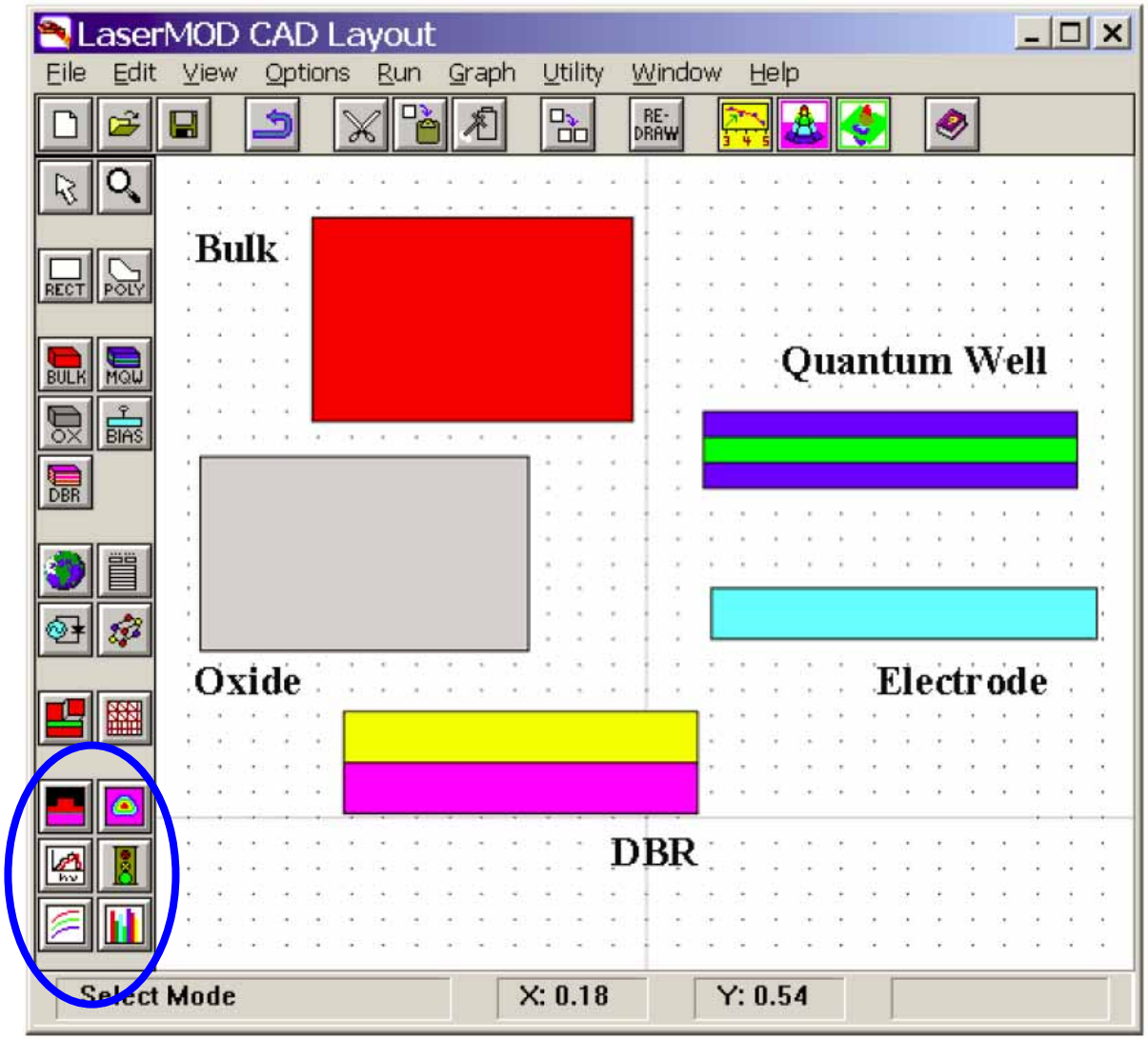
Edit bias points

Edit material parameters





LaserMOD



These are the execute simulation tools



Notes on Tutorial

1) Follow the tutorial in Chapter 4 of the User's Guide.
2) In the material parameter file for AlGaAs, `kpmat_DIELOPT` should be set to `Diel_Opt0`. This affects the index profile.
3) Frequency response must be run after a successful impulse response using the transient bias feature. The impulse response should show the impulse returning to steady-state (as shown in tutorial).
4) Power output for L-I-V curves are for both facets total.
5) In transient, 1st number is time step, 2nd number is # of steps. Make sure all time steps in all transient steps are the same value.
6) Variables are case-sensitive



Notes (continued)

7) If you see multiple lines when viewing graphs with WinPlot, they are lines from previous simulations. To clear the old plots, delete the two directories formed in the same directory as the saved `.las` structure file and run the simulation over.

8) Sheet carrier density (cm^{-2}) = Carrier density (cm^{-3}) x total quantum well thickness (cm)



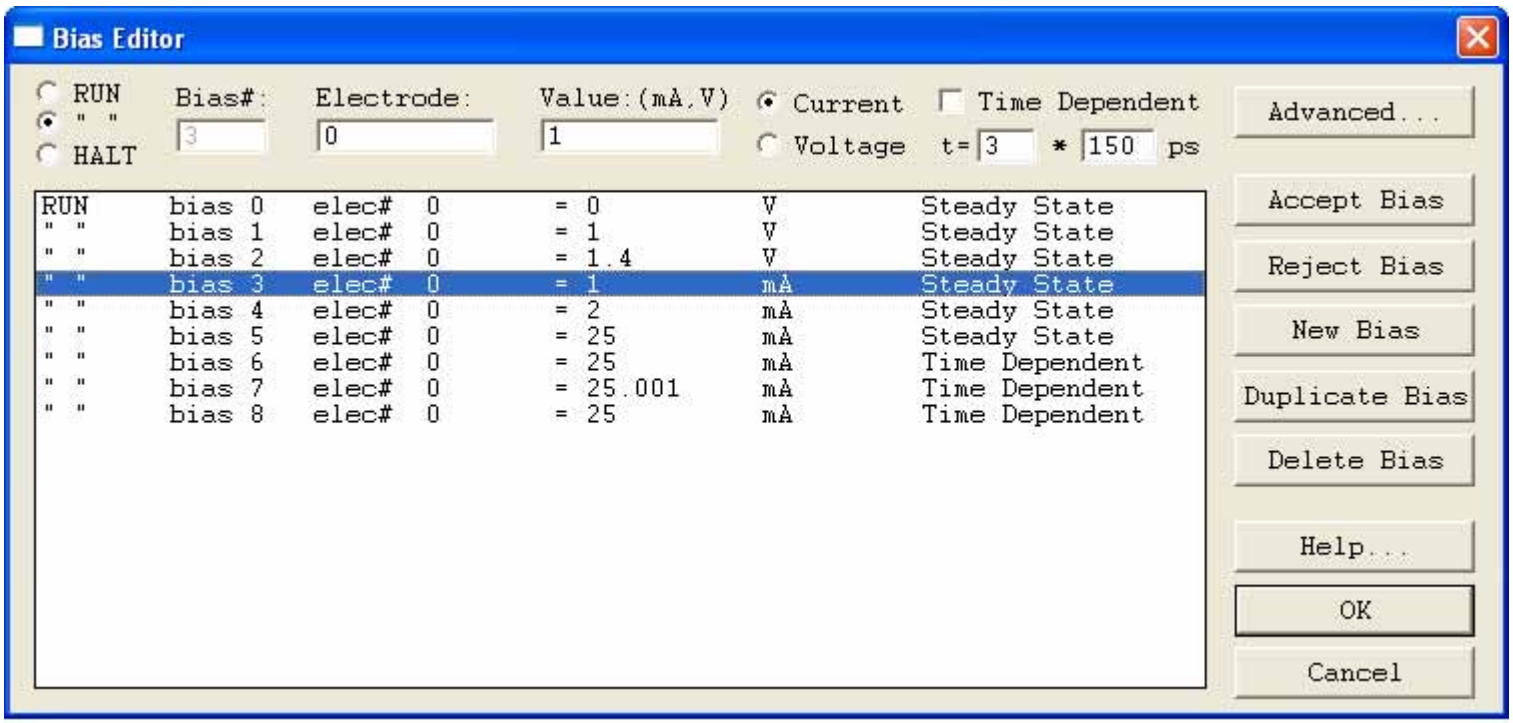
Where to find things

- Simulation parameters can be found in several places:
  - Bias Table (I, V)
  - Global Settings (L, T, α_i, R)
  - Symbol Table (w)
  - Right-clicking on different laser sections (doping, ternary/quaternary element concentrations, w, d)
- Derived parameters can be found in a few graphs:
  - Gain Calculation (g)
  - Simulate Laser output graph (g(v,N), L(I), V(I), transient)
  - Generate Plot output, standard plots (N, IV, LI, freq. resp., etc.)
  - Generate Plot output, custom plots (las. freq., Γ, n_{eff}, etc.)



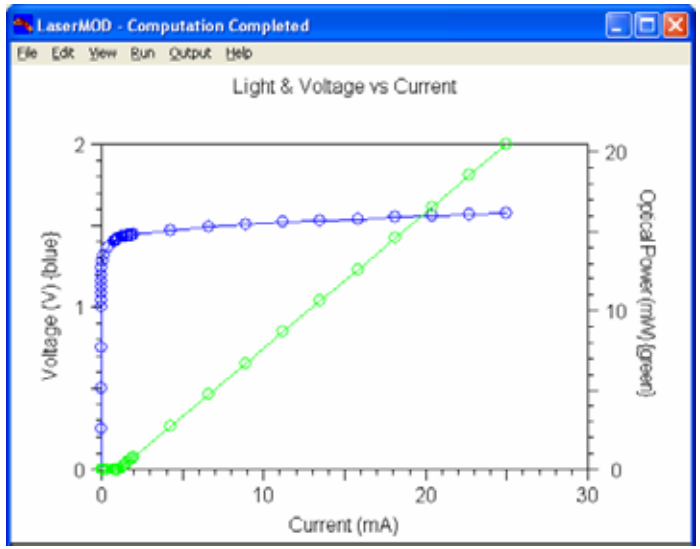
Deviation from Tutorial

- **The manual and program deviate when trying to simulate the IV characteristics, due to updated material files (and a non-updated manual). If the LIV doesn't look right, add bias conditions near threshold, between 1.4V and 25mA. (maybe 1mA and 2mA as shown below.**

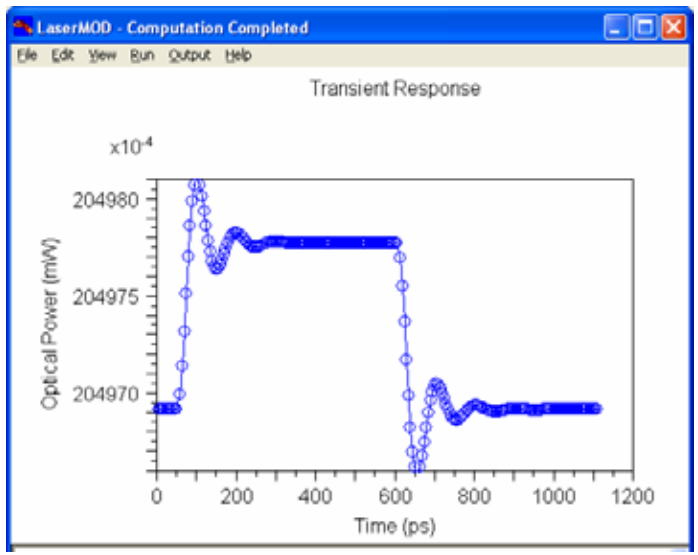


L-I-V and Transient Response with new bias points

- **L-I-V curves should look like this:**



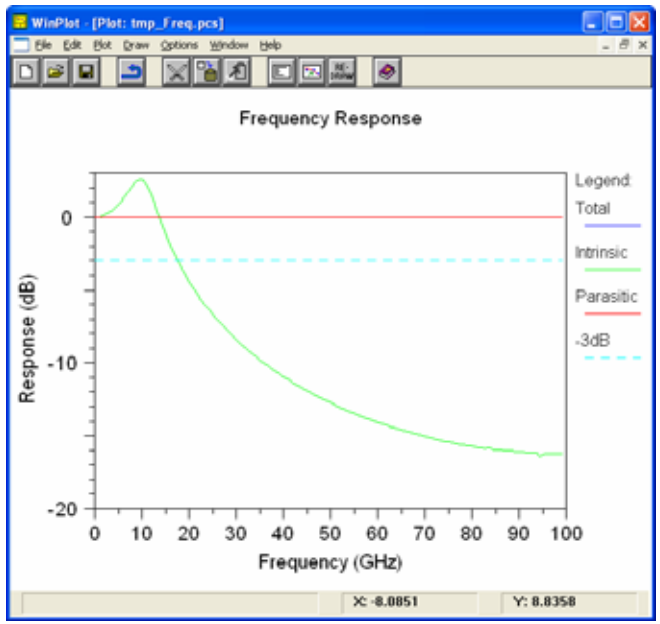
- **Transient response should look like this:**



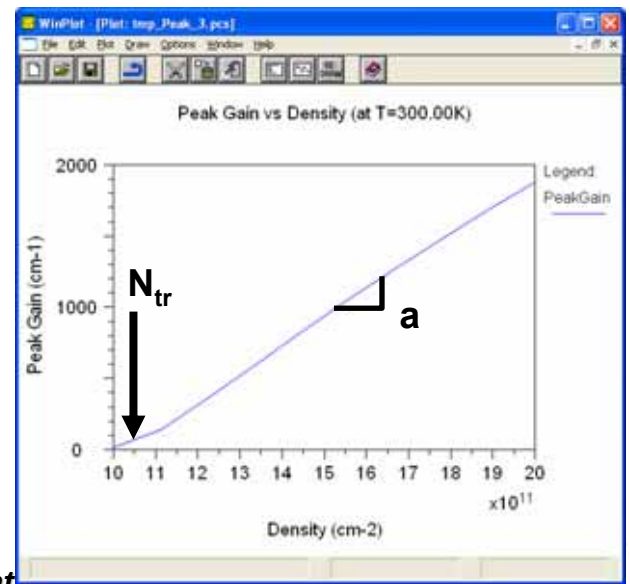


Charts (cont'd) with new bias points

- **Frequency response should look like this:**



- **You can derive differential gain and transparency carrier density from the Peak Gain curve**





Numerical Conditions

- After the tutorial is completed, you should be able to extract several parameters (this is an exercise):

Parameter	Sym.	Approx. Value	Unit	Extracted from…
Threshold current	I_{th}	1	mA	peak gain curve
Differential quantum efficiency	η_d	0.8	W/A	equation
Threshold voltage	V_{th}	1.42	V	LIV curve
Transparency carrier density	N_{tr}	$1.31x10^{18}$	cm^{-3}	peak gain curve
Threshold carrier density	N_{th}	$\sim 3x10^{18}$	cm^{-3}	electron/hole density curve (above threshold)
Differential gain	a	$1.6x10^{-15}$	cm^2	peak gain curve
Confinement factor	Γ	1.88	%	confinement curve
Distributed mirror loss	α_m	20	cm^{-1}	reflectivity value + equation
Lasing frequency	ν	1.24	eV	lasing frequency curve
Effective index	n_{eff}	3.09	-	eff. Index curve
Cavity length	L	500	µm	simulation parameter
Cavity width	w	2	µm	simulation parameter
Quantum well thickness	d	80	Å	simulation parameter
Mirror reflectivity	R	0.37	-	simulation parameter
Threshold gain	g_{th}	1330	cm^{-1}	equation



Useful Formulas

(See Coldren/Corzine, Chapter 2)

- **Output power** $$P_0 = \eta_i \left(\frac{\alpha_m}{\alpha_m + \alpha_i} \right) \frac{h\nu}{q} \left(I - I_{th} \right)$$

- **Mirror loss** $$\alpha_m = \frac{1}{L} \ln\left(\frac{1}{R} \right)$$

- **Threshold gain** $$\Gamma g_{th} = \alpha_i + \alpha_m$$

- **Threshold current** $$I_{th} = \frac{qV}{\eta_i} \left(\frac{N_{th}}{\tau_n} \right) \qquad V = L \cdot w \cdot d$$ **(volume)**

- **Differential quantum efficiency** $$\eta_d = \left(\frac{\alpha_m}{\alpha_m + \alpha_i} \right) \left(\frac{h\nu}{q} \right)$$ **(unitless if without hv/q)**

(Assume internal quantum efficiency, η_i=1)



Things to try

- Obtain the frequency response at a lower bias (I=3 mA) (should be around 2-3 GHz)
- Reduce laser length (L) and observe threshold and η_d increase
- Change # quantum wells, observe threshold change
- Change width, observe threshold change, η_d should stay same
- Change ambient temperature, observe LIV change, wavelength change
- Change mirror reflectivity, observe threshold change and η_d change